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Does common institutional ownership mitigate hold-up problems along the supply chain?

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Abstract

We show that common institutional ownership (CIO) along the supply chain mitigates hold-up problems faced by supplier–customer relationships resulting from incomplete contracts. Suppliers make more relationship-specific investments (RSIs) measured by R&D and patent filings toward their customers that share common institutional investors. Such effect is stronger as the CIO network between a supplier and customer pair becomes wider and deeper. We establish causality by exploiting exogenous shocks to CIO using a broad sample of mergers between financial institutions and further find the CIO effects on suppliers' innovation specificity are stronger for those who ex ante face severer hold-up concerns. Lastly, we provide evidence that CIO involvement increases the combined valuations of supply chain pairs (mainly for customers). Our work sheds light on the hold-up mitigation effect of CIO on firms' decision to make RSIs along the supply chain.

KEYWORDS
common ownership, innovation, investment specificity, supply chain hold-up problems

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1 | INTRODUCTION

Supply chain relationships constitute one of the most important linkages in the economy. Suppliers not only serve as production input providers to their customers, but they are increasingly engaging in R&D and innovation activities that are beneficial to the customers (Cen & Dasgupta, 2021). However, supplier–customer relationships are generally governed by contracts and the challenge of writing complete contracts exposes suppliers to postcontractual opportunistic behaviors by their customers; for example, customers may threaten to walk away from the deal, leaving suppliers' specialized investments much less valuable (Klein et al., 1978). Existing literature has explored solutions to hold-up problems and shown that complete vertical integration between the two parties (Frésard et al., 2020; Seru, 2014), partial equity ownership in the supplier firm by the customer (Fee et al., 2006), and social connections between directors and senior managers of supplier and customer firms (Dasgupta et al., 2021) can mitigate hold-ups and promote relationship-specific investments (RSIs) between suppliers and customers. While previous studies focus on strategic actions taken by the two parties of the bilateral business relationship, little is known about whether a third party outside the supply chain can help overcome this hold-up problem. We fill this gap by drawing insights from a type of third parties that are becoming increasingly prominent among US public firms—common institutional investors.

The hold-up problem arises when a supplier makes RSIs for a customer, including customized production technology, tailored inventory management systems, collaborative R&D projects, dedicated infrastructures, and employee training programs (Crawford, 1990). In many cases, supply chains stand to gain from deeper relationships through RSIs and the exchange of information, owing to the potential for generating relationship rents. However, strong buyers may take advantage of weak suppliers (Goldberg, 1976; Klein et al., 1978), as the value of these investments is much lower outside the relationship, and the relationship could terminate after suppliers have made substantial RSIs (Williamson, 1979). Moreover, there is considerable uncertainty regarding the outcomes of RSIs, making them difficult to describe ex ante and therefore not easily contractible (Tirole, 1999). This uncertainty exacerbates the risk of opportunistic behaviors that favor one party but harm the other, thus undermining the bond between suppliers and customers. Consequently, a supplier facing higher opportunistic risk may strategically underinvest in customer-specific projects, resulting in suboptimal outcomes (Grossman & Hart, 1986).

We hypothesize that common institutional ownership (CIO) can help align interests and reduce opportunistic risks along the supply chain. An institutional investor aims to maximize the risk-adjusted returns of its portfolio rather than individual firm profits. As a result, portfolio diversification can mitigate market frictions (Admati et al., 1994; Clayton & Jorgensen, 2005; Hansen & Lott, 1996; Rubin, 2006). Hansen and Lott's (1996) model suggests that individual firms may impose various types of externalities—pecuniary or nonpecuniary, vertical or horizontal—on one another, and diversification helps internalize these externalities to maximize portfolio returns. Therefore, in supply chain relationships, while a customer can strategically choose to act opportunistically at the expense of its supplier by imposing negative externalities, a customer whose owners also hold stakes in its suppliers is more likely to consider the impact of the externalities on its supplier. As a result, holding stakes in both customers and suppliers could align their incentives and reduce the risk of opportunistic behavior, resulting in an improved coordination with more efficient RSIs and higher overall returns.

One empirical challenge in studying hold-up problems is to reliably measure and quantify investment specificity and the direction toward whom the specialized investments are tailored. In this paper, we use patent cross-citations between supplier and customer firms to construct measures of RSIs, as the citing and cited patents can be linked to individual filing firms, which allows us to use them as a proxy for the citing firm's direction of innovation investments. Our measure is more direct compared to other metrics used in the literature such as patent technology class overlap (Dasgupta et al., 2021) and computational textual analysis of patents (Bena et al., 2021), as these measures cannot identify the direction of innovation investments between suppliers and customers. In addition, innovation investment is particularly relevant to hold-up problems, as innovation investment is a long-term and risky process by nature, rendering it impossible to specify all the contingencies in a contract. More broadly, as a special type of corporate investment, innovation is critical for a firm's long-term competitive advantages and sustainable growth (Porter, 1992).

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To empirically test our hypotheses on CIO as a potential solution to hold-up problems, we extract a large sample of supplier–customer pairs from the Compustat Segment Files and obtain institutional holdings data from Thomson's CDA/Spectrum database (form 13F). Using a pairwise fixed-effect model, we first exploit within supplier–customer pair variation in CIO and show that suppliers conduct more R&D when they and their major customers are commonly held by institutional investors. Next, we show that higher R&D expenditures translate into more innovation outputs, as measured by number of patents filed and eventually granted to suppliers. We then use patents' cross-citations between supplier and customer firms to evaluate the specificity of innovation investments. Our results indicate that suppliers produce more innovation outputs that are specific to their major customers when they share common ownership linkages with their customers. The above findings are robust to controlling for suppliers' and customers' overall institutional investors' holdings levels, in addition to an array of firm and firm-pair level characteristics as well as year fixed effects.

Nevertheless, institutional investors conduct dedicated research and analysis to pick their investment targets. The nonrandom nature of the investment decision process leads to an endogeneity concern that investors may choose to hold supplier and customer firms with certain innovation prospects. Moreover, unobservable firm characteristics may also correlate with both our explanatory and outcome variables, thus making the ordinary least square (OLS) results spurious. To address these endogeneity concerns, we exploit a quasi-natural experiment of financial institution mergers using a difference-in-differences (DiD) framework (He & Huang, 2017; Lewellen & Lowry, 2021). When two financial institutions merge, a firm held by one merging institution is likely to experience an increase in CIO when one of its major customers is held by the other merging institution prior to the merger.[1] Thus, the treatment sample consists of firms whose ownership linkages with customer firms are likely to increase just because of the merger, while the control sample consists of other firms in the same institution's portfolio that are unlikely to experience such changes. We find evidence that treatment firms, relative to control firms, make more relationship-specific innovation investments in the year after the institution mergers, which suggests a causal impact of CIO on suppliers' specialized innovation investments made toward their customers.

Our DiD analysis reveals that treatment firms, compared to control firms, exhibit a higher propensity for making specialized innovation investments targeted toward their commonly held customers in the year following institutional mergers. This finding corroborates our OLS results and helps us establish a causal relation between CIO and suppliers' innovation specificity toward their customers. Notably, these results remain robust across various model specifications, including the incorporation of merger-time pair-fixed effects as well as merger- and pair-fixed effects. The results are also robust when we look at innovation investments in the post-3-year window and when accounting for heterogeneous treatment effects in DiD settings.

Building upon this DiD framework, we further investigate heterogeneous effects to provide additional evidence supporting the mitigation of hold-up concerns as the underlying mechanism driving the increased innovation specificity. If CIO indeed fosters supplier innovation specificity by alleviating hold-up concerns, we expect to observe a more pronounced increase in investment specificity among suppliers facing more severe hold-up concerns ex ante. We use two measures to capture the severity of the hold-up problems. The first measure is asset redeployability, developed by Kim and Kung (2017), which accounts for the usability of assets within and across industries. More redeployable assets exhibit higher recovery rates and are traded more actively in secondary markets. For suppliers with higher asset redeployability, they should be less concerned about being held up by their customers, since it is relatively easier for these suppliers to find other buyers in case of a hold-up. Our second measure is the Herfindahl–Hirschman Index (HHI), which serves as a proxy for suppliers' and customers' relative bargaining powers. For suppliers operating in industries with higher HHI, an indicator of less competition within the industry, they should possess higher bargaining power relative to their customers, and thus have less hold-up concerns. On the other hand, if customers hold more bargaining power, suppliers should be more concerned about being held up in the future. We find that the increase in suppliers' innovation specificity is lower after institution mergers for suppliers with higher asset redeployability or

[1] Our definition of held is that greater than 1% of outstanding shares are owned by the institution.

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greater bargaining power, while such increase is higher when customers hold more bargaining power. In other words, suppliers tend to allocate more resources to RSIs following a positive shock to their CIO linkages with customers if suppliers have higher hold-up concerns ex ante.

Next, we investigate and rule out some alternative explanations for our main results. In particular, we examine whether knowledge spillovers and innovation collaboration between suppliers and customers are the channels leading to increased supplier innovation specificity. If knowledge spillovers were indeed the primary mechanism, we would anticipate observing effects on customers' innovation specificity. In this scenario, customers' innovation endeavors would become more tailored toward their suppliers, indicating a convergence of innovation efforts toward a common direction. This alignment would result from the exchange of ideas and know-how, facilitating a shared knowledge reservoir between suppliers and customers. However, our results show that customers' innovations exhibit reduced specificity toward their suppliers following institutional mergers. This discrepancy contradicts the notion that knowledge spillovers are the driving force behind the increase in supplier innovation specificity.

Then, we investigate whether firm-pairs initiate or terminate joint ventures and strategic alliances after becoming commonly owned, as these are significant organizational mechanisms for fostering innovation collaboration (Robinson, 2008; Seru, 2014). Our analysis shows that treated firm-pairs are less inclined to establish an innovation collaboration entity after the institution merger.

He and Huang (2017) also examine joint ventures and strategic alliances using the institution merger setting. They find that cross-ownership improves firms' tendency to engage in joint ventures, strategic alliances, or acquisitions. While it may seem that we provide an opposite finding using the same setting, it is important to note that their study focuses on the cross-ownership of same-industry firms, whereas our study examines the cross-ownership of supply chain partners. Johnson and Houston (2000) find that horizontal joint ventures create synergistic gains for both partners, while vertical joint ventures generate gains only for suppliers. Supplier–customer pairs only choose joint ventures when hold-up hazards are high due to a governance motive, as firms become mutually reliant on a venture framework to reduce potential hold-up hazards. Given that our baseline results suggest that CIO among supply chain partners mitigates hold-up problems, our findings corroborate this line of research by demonstrating the substitutive effect of CIO on collaboration in terms of joint ventures and strategic alliances.

As the last step of our analyses, we examine the welfare implications of CIO on the value of the suppliers and customers. We investigate this effect at both the firm and firm-pair levels, specifically on the market valuations of suppliers, customers, and joint parties. Our findings reveal that the combined market value of supplier and customer pairs increases following CIO involvement. At the firm level, we find that the substantial increase in market value is concentrated among customers rather than suppliers. In addition, we show that the effect of our main findings is stronger for active investors rather than passive investors. These results are consistent with our predictions and elucidate investors' incentives for promoting RSIs made from suppliers toward their commonly held customers, as suppliers stand to benefit from such coordination while customers can benefit from the value effects. Our findings suggest that CIO involvement can lead to Pareto improvements in such scenarios, resulting in advantageous outcomes for the three parties involved.

Our paper is related to several strands of literature. First, we add to the literature on the interaction between transaction cost economics and supply chain management by showing that CIO along the supply chain can potentially mitigate hold-up problems. Williamson (1979) and Grossman and Hart (1986) develop the transaction costs theory and property rights theory, respectively, to tackle the questions of whether a firm's transactions with other firms should remain at arm's length or internalized via vertical integration, when RSIs are needed, but faced with incomplete contracts and hold-up problems. Klein et al. (1978) argue that complete vertical integration can mitigate hold-up problems in the supply chain, so that mergers along the supply chain can lead to increased efficiency (see also Frésard et al., 2020; Seru, 2014). Fee et al. (2006) show that customer's equity stake in its supplier firm also encourages efficient actions when firms are concerned with expropriation or hold-up problems arising from RSIs. A more recent study by Dasgupta et al. (2021) finds social connections between directors and senior managers of supplier and customer firms help mitigate hold-up concerns. Prior studies focus on strategic actions taken by the firms themselves to solve the

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hold-up problems. Our paper contributes to this literature by uncovering a third party, that is, common institutional investors, that also helps to mitigate hold-up problems along the supply chain and promote RSIs.

Our paper also contributes to the burgeoning literature of CIO by looking at the cross-held firms along the supply chain. The literature has documented that as common owners are incentivized to internalize each firm's actions to maximize the portfolio benefit (Hansen & Lott, 1996; Rubin, 2006), CIOs could have real impacts on firms' strategic choices, including product market competition (Azar et al., 2018; Dennis et al., 2022; He & Huang, 2017; Koch et al., 2021; Lewellen & Lowry, 2021; López & Vives, 2019), managerial incentives and compensations (Antón et al., 2023; Gilje et al., 2020; Liang, 2016), corporate governance (He et al., 2019), executive, and financial disclosure and accounting accruals (He et al., 2020; Jung, 2013; Park et al., 2019; Pawliczek et al., 2022). The literature began by focusing on horizontal CIOs (i.e., commonly owned firms in the same industry) (Azar et al., 2018; He & Huang, 2017; He et al., 2019; Koch et al., 2021; López & Vives, 2019; Park et al., 2019), and more recently, moved on to examine CIOs between firms from different industries that are economically linked in different contexts, such as CIOs between acquirers and target firms (Brooks et al., 2018; Harford et al., 2011; Matvos & Ostrovsky, 2008), banks and industrial firms (Azar et al., 2022; He et al., 2023; Ojeda, 2019; Wang & Wang, 2022), media and industrial firms (Chen et al., 2022; He et al., 2022, Ru et al., 2020). While growing fast, the literature has paid little attention to the effects of CIO along supply chain firms. Our paper is among the first to extend this strand of literature.

While Freeman (2023) and Cheung et al. (2020) also examine CIO effects on supply chain issues, our study distinguishes itself in several ways. First, Freeman (2023) focuses on the supply chain partnership duration and Cheung et al. (2020) focus on suppliers' operating performance. In contrast, our focus lies on the specificity of supplier innovation investments tailored to their customers. Second, we employ institution mergers as a quasi-natural experiment to provide causal inference, while Cheung et al. (2020) primarily demonstrate correlations. While Freeman (2023) employs a similar DiD approach, our methodology involves supplier–customer pairwise fixed effects rather than supplier- and customer-fixed effects. This enables us to account for changes in supplier innovation activities directed at specific customers within a given relationship, thus addressing supply chain hold-up concerns more directly. Third, our sample spans from 1980 till 2019 while Freeman's stops in 2010 and Cheung et al.'s in 2011, making the results and implications of our study applicable to a wider and more recent time period.

Our paper also adds to the literature on corporate innovation. A vast body of literature investigates various factors that foster or hinder innovation, including financial market developments (Hsu et al., 2014; Moshirian et al., 2021), shareholder characteristics (Aghion et al., 2013; Brav et al., 2018; Chemmanur et al., 2014), managerial incentives (Manso, 2011), CEO traits (Hirshleifer et al., 2012), financial analysts (He & Tian, 2013), firm boundaries (Seru, 2014), and product market competition (Aghion et al., 2005).[2] Fewer studies have looked at innovation activities through the lens of institutional cross-ownership. He and Huang (2017) find that cross-ownership enhances firms' innovation productivity. Kostovetsky and Manconi (2020) find that CIO facilitates the diffusion of innovation across firms. Using project-level data, Li et al. (2023) find that CIOs prompt venture capitalists to terminate underperforming drug projects and withhold funding from struggling startups. Antón et al. (2024) show that common ownership decreases innovations since common owners are incentivized to discourage business stealing between cross-held firms that compete in one product market, while common ownership can only increase innovation when technological spillovers are sufficiently large relative to product market spillovers. Compared to prior studies that focus on knowledge spillovers and innovation outcomes in general, we are interested in hold-up mitigation and we show that CIO can help reduce the supply chain hold-up costs. Furthermore, we provide suggestive evidence that RSIs induced by CIO can substitute for formal collaborations, such as joint ventures and strategic alliances, between suppliers and customers.

[2] See He and Tian (2020) for a comprehensive review on the innovation literature.

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2 | DATA, SAMPLE, AND VARIABLES

2.1 | Data sources

Our supplier–customer relationship sample is from Compustat Segment Files, covering firm-years from 1978 to 2019. We stop at the end of 2019 to avoid the impact of the COVID-19 pandemic on firms' activities. According to the Statement of Financial Accounting Standards (SFAS) No. 131, firms are required to disclose the names of customers that account for more than 10% of their total sales, while some firms voluntarily report customers even below this threshold. We exclude government buyers or generic customers such as "Foreign Sales," "Major Customer," "Vendor," or "Not Reported" in our sample. Since the Compustat Segment database only discloses customer names, and in many cases, only name abbreviations, it requires manual matching of each disclosed name to a Compustat identifier. We use the supplier–customer links provided by Barrot and Sauvagnat (2016), who use a string-matching algorithm to match customer names to their corresponding firms in Compustat from 1978 to 2013, and augmented the data coverage to 2019 with a fuzzy name-matching algorithm between company names from Compustat Segment Files and names in Compustat/CRSP.

Data on institutional ownership is from the Thomson (13F) database. This database includes institutional investors with over $100 million in assets under management, including pension funds, endowments, insurance companies, bank trusts, mutual funds, hedge funds, and independent advisors. The ownership data report institutions' holdings at the end of each calendar quarter. Institutional investors' stock holding information from the Thomson (13F) database is available from 1980 to the present. We then merge Thomson (13F) data with CRSP's quarterly database to obtain data on firms' shares outstanding and share prices.

Finally, we require both supplier and customer firms to have nonmissing relevant financial information to be included in our final sample. We also require firm-year observations to have nonnegative values for total assets and sales. Firm-level information is obtained from the Compustat/CRSP merged database. Following prior supply chain studies (Banerjee et al., 2008, 2021), we exclude financial firms (Standard Industrial Classification [SIC] code 6000–6999) and utility firms (SIC codes 4900–4999) as they are subject to different sets of regulatory requirements. The final sample covers the periods from 1980 to 2019, consisting of 22,847 supplier–customer pair by year observations with 2715 unique suppliers, 983 unique customers, and 4805 unique supplier–customer pairs.

2.2 | Investment specificity measures

We are interested in how CIO influences suppliers' specialized investments made toward customers. We investigate three dimensions of investments on innovation activities: innovation input, innovation output, and innovation specificity, with most attention devoted to innovation specificity. R&D expenditures have been widely used in the literature to proxy for innovation input (Allen & Phillips, 2000). Specifically, we scale R&D expenses by the book value of total assets and replace them with zero if R&D is missing. For innovation output, we count the number of patents filed and eventually granted in each application year (Fang et al., 2014). We retrieve patent data from Kogan et al. (2017), which complies the entire universe of US patents from 1926 to 2020.[3] The primary interest of this paper is innovation specificity, as RSI is the focal issue in hold-up problems. Kelly et al. (2021) further provide patent pairwise citations for all United States Patent and Trademark Office (USPTO) patents from 1836 to 2022.[4] We construct two pairwise measures based on their cross-citations data to capture whether suppliers' innovative activities are more tailored to the

[3] This data set includes basic information on patent number, filing date, issue date, CRSP permco, and patent values. It can be accessed via https://github.com/KPSS2017/Technological-Innovation-Resource-Allocation-and-Growth-Extended-Data.

[4] This data set includes information on patent similarities, importance, and pairwise cross-citations. It can be accessed via https://github.com/KPSS2017/Measuring-Technological-Innovation-Over-the-Long-Run-Extended-Data/blob/main/README.md.

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needs of their major customers (Alcacer & Gittelman, 2006; Jaffe et al., 2000). The first measure is *TotalCiter*, the number of supplier patents that cite their customer's patents in an application year; the second measure is *TotalCited*, the number of customer patents that cited by their supplier in an application year. These two measures together capture the width and depth of the extent to which suppliers tailor their R&D activities toward their major customers. If more supplier patents cite their major customer's patents, and cite a more diverse range of customer patents, we can interpret this as the supplier conducting more relationship-specific innovation investment toward its customer. These cross-citation measures can deliver more direct information about investment specificity compared with other measures employed by the literature, such as patent technology classes overlap (Dasgupta et al., 2021) and computational textual analysis of patents (Bena et al., 2021). Both these two measures aim to capture the similarity of the patents' technology, which should naturally be high for firms along the supply chain that mostly operate in the same industry. However, similarity alone provides no information about direction, which is our main focus in the paper. By employing cross-citation information, we can directly quantify whether suppliers tailor their innovation investment toward their customers.

2.3 | CIO measures

Following He and Huang (2017), we construct two measures of CIO, *NumCross* and *TotalCrossOwn*. *NumCross* is defined as the number of unique common investors holding both the supplier and customer (with holdings larger than 1% of total shares outstanding). This measure is discrete, capturing the width of common investor networks. *TotalCrossOwn* is calculated as the sum of supplier shares held across common investors, divided by the supplier's total share outstanding (also considering only holdings larger than 1% of total shares outstanding). This measure captures the level of influence that common investors can exert on suppliers, since we are interested in whether common ownership can incentivize suppliers to conduct more relationship-specific R&D. We calculate the CIO measures for each quarter and then average to construct an annual figure. In all specifications, CIO is measured with a 1-year lag.

2.4 | Control variables

Following prior literature (Banerjee et al., 2008; Fee et al., 2006), we construct and control for an array of supplier, customer, and firm-pair-level characteristics in the regression analyses. We control for the same set of characteristics for both suppliers and customers, including institutional ownership (sum of total percentage institutional holdings reported on 13F), firm size (natural logarithmic of the book value of total assets), leverage (total debt divided by the book value of total assets), market-to-book ratio (market value of total assets divided by the book value of total assets), asset tangibility (net property, plant, and equipment [PPE] divided by the book value of total assets), return on assets (operating income divided by the book value of total assets), and capital expenditures (capital expenditures divided by the book value of total assets). To take into consideration pairwise strength and relative importance of business relationship, we control for pairwise sales fraction (supplier's sales to each major customer divided by total sales).

2.5 | Summary statistics

We tabulate the summary statistics of the main variables in the empirical analysis in Table 1. On average, a supplier–customer pair in our sample has 2.1 common investors, which confirms that US public firms indeed are highly interconnected through institutional investors. Common investors on average hold 7.3% of supplier shares, thus they have certain level of power affecting supplier decisions or have enough stakes in the business relationship to care about being held up. Moreover, in our DiD sample, a supplier–customer pair is commonly held by the merging

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TABLE 1 Summary statistics.

	N	Mean	SD	P25	P50	P75
Dependent variables						
Sup_R&D	22,874	0.080	0.159	0.000	0.021	0.103
ln(Patents+1)	22,874	0.885	1.400	0.000	0.000	1.386
TotalCiter	8783	2.202	7.528	0.000	0.000	1.000
TotalCited	8783	2.912	9.687	0.000	0.000	1.000
Sup_TobinQ	4887	2.043	1.886	1.199	1.574	2.253
Cust_TobinQ	4887	1.965	1.338	1.164	1.583	2.170
Pair_TobinQ	4887	1.972	1.130	1.236	1.676	2.206
Alliances	4887	0.015	0.120	0.000	0.000	0.000
Independent variables						
NumCross	22,874	2.107	2.348	0.000	1.250	3.500
TotalCrossOwn	22,874	0.073	0.097	0.000	0.028	0.114
Redeploy	4887	0.369	0.070	0.337	0.388	0.416
Sup_HHI	4887	0.304	0.226	0.124	0.222	0.413
Cust_HHI	4887	0.393	0.222	0.227	0.346	0.545
Control variables						
Sup_InstOwn	22,874	0.338	0.219	0.140	0.334	0.533
Sup_Size	22,874	5.512	2.104	3.956	5.322	6.946
Sup_Leverage	22,874	0.213	0.225	0.015	0.143	0.338
Sup_MB	22,874	1.990	4.117	1.089	1.459	2.179
Sup_Tangibility	22,874	0.262	0.221	0.092	0.195	0.370
Sup_ROA	22,874	0.062	0.254	0.034	0.110	0.166
Sup_Capex	22,874	0.063	0.169	0.025	0.049	0.090
Cust_InstOwn	22,874	0.405	0.170	0.301	0.430	0.531
Cust_Size	22,874	9.975	1.835	8.943	10.185	11.332
Cust_Leverage	22,874	0.265	0.230	0.102	0.190	0.364
Cust_MB	22,874	1.883	1.264	1.135	1.491	2.113
Cust_Tangibility	22,874	0.319	0.205	0.138	0.281	0.485
Cust_ROA	22,874	0.138	0.101	0.091	0.138	0.178
Cust_Capex	22,874	0.066	0.063	0.030	0.056	0.091
SalesFraction	22,874	0.186	0.166	0.097	0.140	0.225

institution after the merger for an average of 5.5 years, which suggests that the effect of CIO lasts long enough for innovation outcomes to be realized.

On average, a supplier invests 8% of total assets in R&D, while a customer only invests 3.4% of total assets in R&D, suggesting that R&D intensity is much larger for suppliers than customers. This is consistent with Dasgupta et al. (2021), which confirm that supplier innovation is an important part of innovation along the supply chain, and for the economy as a whole. On average, a customer is much larger than a supplier in terms of asset size, with average supplier size only 1.15% of customer size, consistent with the findings in Fee et al. (2006) that disclosed customers are the principal ones. Suppliers and customers in our sample both have high levels of institutional ownership, with average

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institutional ownership of 33.8% for suppliers and 40.5% for customers. Customers are more leveraged than suppliers, with average debt-to-total assets ratio of 26.5% for customers and 21.3% for suppliers. On average, a supplier sells 18.6% of its total sales to a major customer. Thus, suppliers heavily depend on their major customers.

For patenting variables, on average, a supplier files for (and is eventually granted) 18 patents in an application year and cites three customer's patents in two new patent applications.

3 | EMPIRICAL METHODOLOGY AND RESULTS

3.1 | OLS analysis: Innovation quantity

We begin our analyses by investigating whether CIO promotes suppliers to make more innovation investments in general. We regress supplier innovative activities on the pairwise common ownership measures with the following regression specification:

$$Sup_Innovation_{i,t+1} = \alpha CIO_{i,j,t} + \beta X_{i,t} + \gamma Y_{j,t} + \delta Z_{i,j,t} + \mu_{i,j} + \upsilon_t + \varepsilon_{i,t+1}, \quad (1)$$

where i, j, and t denote supplier, customer, and year, respectively. $Sup_Innovation_{i,t+1}$ represents supplier-level innovative activities defined in Section 2.2 (including innovative input, i.e., R&D expenditures, and innovative output, i.e., patenting outcomes). We measure the dependent variable in year $t + 1$ since it takes time for innovative ideas to materialize. $CIO_{i,j,t}$ is the main variable of interest (including two different measures) that captures the common investor networks, defined in Section 2.3. In addition, we control for a series of supplier-level ($X_{i,t}$), customer-level ($Y_{j,t}$), and supplier–customer pair-level ($Z_{i,j,t}$) characteristics that might affect supplier innovation. Finally, we include year fixed effects (υ_t) to capture the time trend of innovation activities and possible fluctuation of overall economic conditions, and supplier–customer pair fixed effects ($\mu_{i,j}$) to control for any unobserved time-invariant heterogeneity at the relationship level (e.g., unobserved technology synergy). Supplier–customer-pair fixed effects ensure that our results capture the effect of the within-relationship variations in pairwise CIO on supplier innovation. In other words, any changes in supplier innovation activities can be seen as reactions targeting the customer in the specific relationship pair. We cluster standard errors at the supplier–customer pair level to account for pairwise serial correlation of the residuals.

Table 2 reports the regression results on supplier R&D expenditures and patenting outcomes. In columns (1) and (2), we consistently observe a positive effect of CIO on supplier R&D (significant at the 1% level). For example, column (1) suggests that, when they become commonly held (by one additional common investor) with its major customer, holding all else equal, suppliers' R&D expenditures increase by 0.2% of total assets, a 2.50% increase relative to the sample mean. Column (2) shows that a 10% increase in *TotalCrossOwn*, that is, holding 10% more of the total market values of the supplier–customer pair, increases suppliers' R&D expenditures by 0.5% of their total assets, equivalent to a 6.1% increase relative to the sample mean. Columns (3) and (4) report the regression results on supplier patenting outcomes. Consistent with the results in R&D expenditures, we find a strong correlation between common ownership and supplier innovation output. Column (3) shows that the number of supplier patents increases by 2.2% if a supplier becomes commonly held (by one additional common investor) with its major customer; while column (4) suggests that a one-standard-deviation increase in *TotalCrossOwn* translates into a 0.84% increase in the number of supplier patents, though not statistically significant. Results in Table 2 lend support to our hypothesis that CIO along the supply chain mitigates the underinvestment (in specialized assets) problems, promoting both supplier innovation inputs and outputs.

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TABLE 2 Common ownership and supplier R&D and innovation output.

	(1)	(2)	(3)	(4)
	Sup_R&D		*ln(Patent + 1)*	
NumCross	0.002***		0.022***	
	[0.001]		[0.008]	
TotalCrossOwn		0.049***		0.084
		[0.013]		[0.164]
Sup_InstOwn	0.007	0.006	−0.264***	−0.179**
	[0.008]	[0.008]	[0.091]	[0.088]
Sup_Size	−0.035***	−0.035***	0.176***	0.181***
	[0.003]	[0.003]	[0.024]	[0.024]
Sup_Leverage	−0.034***	−0.034***	−0.254***	−0.260***
	[0.010]	[0.010]	[0.052]	[0.052]
Sup_MB	0.001	0.001	0.002*	0.002*
	[0.001]	[0.001]	[0.001]	[0.001]
Sup_Tangibility	0.049***	0.049***	0.138	0.140
	[0.016]	[0.016]	[0.114]	[0.113]
Sup_ROA	−0.174***	−0.174***	−0.005	−0.005
	[0.040]	[0.040]	[0.029]	[0.029]
Sup_Capex	0.044***	0.044***	−0.040*	−0.043**
	[0.014]	[0.014]	[0.022]	[0.022]
Cust_InstOwn	0.006	0.006	−0.219**	−0.160
	[0.011]	[0.011]	[0.102]	[0.101]
Cust_Size	0.003	0.003	−0.034	−0.035
	[0.002]	[0.002]	[0.036]	[0.036]
Cust_Leverage	−0.006	−0.006	−0.165**	−0.166**
	[0.006]	[0.006]	[0.067]	[0.067]
Cust_MB	0.001	0.001	−0.009	−0.009
	[0.001]	[0.001]	[0.008]	[0.008]
Cust_Tangibility	−0.023*	−0.023*	0.085	0.081
	[0.013]	[0.013]	[0.153]	[0.153]
Cust_ROA	0.020	0.020	−0.139*	−0.147*
	[0.014]	[0.014]	[0.078]	[0.079]
Cust_Capex	0.013	0.013	0.028	0.024
	[0.011]	[0.011]	[0.082]	[0.082]
SalesFraction	−0.003	−0.003	−0.013	−0.012
	[0.011]	[0.011]	[0.050]	[0.051]
Year FE	Yes	Yes	Yes	Yes
Firm-Pair FE	Yes	Yes	Yes	Yes
Observations	22,874	22,874	22,874	22,874
Adj. R^2	0.786	0.786	0.866	0.865

(Continues)

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TABLE 2 (Continued)

Notes: This table examines the OLS relationship between supplier R&D and the level of pairwise common institutional ownership. The dependent variable is supplier R&D expense over the book value of total assets. The main explanatory variables are two different measures of common institutional ownership, *NumCross* and *TotalCrossOwn*, as defined in Appendix A. Supplier level, customer level, and pairwise control variables are defined in Appendix A. Year and firm-pair fixed effects are included in all regressions. Standard errors, reported in brackets, are clustered at supplier–customer pair levels. ***, **, and * denote statistical significance at the 1%, 5%, and10% level, respectively.

3.2 | OLS analysis: Innovation specificity

Having shown that CIO is associated with suppliers conducting more innovation activities, we proceed to investigate whether these innovations are more specialized toward their commonly held customers, which is the focal issue of the hold-up problem. We take advantage of the patent pairwise cross-citations data to gauge suppliers' innovation specificity and estimate Equation (1) by replacing *Sup_Innovation* with *Innovation_Specificity*, which is measured by *TotalCiter*, the number of supplier patents that cite their customer's patents in an application year, and *TotalCited*, the number of customer patents that cited by their supplier in an application year. We tabulate the regression results in Table 3. For the *TotalCiter* measure, *NumCross* and *TotalCrossOwn* have positive and significant coefficients (at the 1% level). Column (1) suggests that, on average, being commonly held by one additional common investor with its major customer is associated with the supplier citing 0.388 more of its commonly held customers' patents, which is equivalent to a 17.6% increase relative to the sample mean. Similarly, in column (4) we show that a 10% increase in the holding of the total market values of supplier–customer pairs will bolster suppliers cite 0.747 more of their commonly held customers' patents, a 25.7% increase relative to the sample mean. Results in this section further support our main hypothesis that common ownership incentivizes suppliers to conduct more relationship-specific innovation toward their commonly held customers by alleviating suppliers' concerns of being held up.

3.3 | Identification

As discussed earlier, an endogeneity concern of our OLS results is that omitted variables correlated with both a supplier being commonly held with its major customers and its innovation prospects could bias the results. Another concern is that supply chains with better innovation synergies or technological cooperation prospects may attract more common institutional investors, resulting in reverse causality. In this section, we address these concerns using an identification strategy that is widely exploited in the CIO literature: a DiD approach based on the quasi-natural experiment of financial institution mergers that generates plausibly exogenous variation of a firm's common ownership levels.[5] Financial institutions usually merge for reasons unrelated to the fundamentals of their portfolio holdings. He and Huang (2017) show that the majority of financial institution mergers happen either due to deregulation or strategic considerations of the investors. In addition, the acquiring investor usually maintains the holdings of the target investor for a period of time following a merger, possibly due to liquidity or transaction cost concerns. Therefore, if one party of the supply chain is held by one of the merging institutions before the merger, while the other party of the supply chain is held by the other merging institution, then both firms would be held (commonly held) by the same institution right after the merger. This event provides us with a nice quasi-natural experiment to test how common

[5] Although being commonly used in the literature, Lewellen and Lowry (2021) show that this identification strategy has some problems in horizontal common ownership studies that have been previously overlooked. Despite that there are several dozens of financial institution merger events, close to half of the treatment firms are associated with mergers occurring during 2008 and 2009, the largest of which is the Blackrock–BGI merger. This means that the postmerger period for these firms coincides with the aftermath of the financial crisis. This problem may be a serious one for studies looking at horizontal effects (firms competing in the same industry), because in these studies, control firms are sampled from different industries than treatment firms. Different industries could have responded differently to the crisis even in the absence of the merger event. This problem is less of a concern for studies on vertical effects (supply chain relationships), because treatment firms and control firms do not have to come from different industries.

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TABLE 3 Common ownership and supplier innovation specificity.

	(1)	(2)	(3)	(4)
	TotalCiter		*TotalCited*	
NumCross	0.388***		3.708	
	[0.116]		[2.302]	
TotalCrossOwn		0.623***		7.473***
		[0.160]		[2.847]
Sup_InstOwn	−4.532***	−7.549***	−3.329**	−5.954***
	[1.483]	[2.036]	[1.320]	[1.810]
Sup_Size	0.943***	1.217***	1.009***	1.320***
	[0.244]	[0.293]	[0.246]	[0.295]
Sup_Leverage	−2.614***	−3.161***	−2.722***	−3.327***
	[0.949]	[1.114]	[0.957]	[1.132]
Sup_MB	0.002	0.000	0.002	0.000
	[0.002]	[0.003]	[0.002]	[0.003]
Sup_Tangibility	1.433	2.921	1.315	2.718
	[1.880]	[2.188]	[1.870]	[2.190]
Sup_ROA	−0.613*	−0.907*	−0.587*	−0.873*
	[0.350]	[0.484]	[0.350]	[0.486]
Sup_Capex	1.589**	0.928	1.638**	1.021
	[0.804]	[1.083]	[0.798]	[1.083]
Cust_InstOwn	−1.370	0.639	−0.373	2.039
	[1.736]	[2.258]	[1.756]	[2.300]
Cust_Size	1.092**	1.401**	1.045**	1.328**
	[0.487]	[0.600]	[0.487]	[0.604]
Cust_Leverage	1.757	2.829*	1.801	2.903*
	[1.262]	[1.538]	[1.282]	[1.571]
Cust_MB	0.235	0.254	0.238	0.258
	[0.143]	[0.162]	[0.145]	[0.164]
Cust_Tangibility	2.918	4.483	2.692	4.159
	[2.137]	[3.043]	[2.154]	[3.107]
Cust_ROA	−0.789	−1.602*	−1.013	−1.945**
	[0.736]	[0.868]	[0.757]	[0.908]
Cust_Capex	−1.364	−1.421	−1.119	−1.049
	[1.491]	[2.276]	[1.510]	[2.340]
SalesFraction	−0.134	−0.225	−0.080	−0.145
	[0.387]	[0.541]	[0.387]	[0.542]
Year FE	Yes	Yes	Yes	Yes
Firm-Pair FE	Yes	Yes	Yes	Yes
Observations	8783	8783	8783	8783
Adj. R^2	0.653	0.656	0.651	0.653

(Continues)

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TABLE 3 (Continued)

Notes: This table examines the OLS relationship between supplier innovation specificity and the level of pairwise common institutional ownership. The dependent variables, *TotalCiter* and *TotalCited*, are different measures of supplier innovation specificity, which are defined in Table A1. The main explanatory variables are two different measures of common institutional ownership, *NumCross* and *TotalCrossOwn*, as defined in Appendix A. Supplier level, customer level, and pairwise control variables are defined in Appendix A. Year and firm-pair fixed effects are included in all regressions. Standard errors, reported in brackets, are clustered at supplier–customer pair levels. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

ownership along the supply chain affects supplier innovation specificity. A key advantage of this identification strategy is that there are multiple shocks that affect different firms at exogenously different times. Identification with multiple shocks can alleviate the concerns in studies using only one single shock, namely, potential omitted variables coinciding with the single shock directly affect the outcome of interest.

The financial institution merger events used in our paper are from Lewellen and Lowry (2021), which includes a total of 64 merger events from 1984 to 2010. Given the merger events available, we keep the DiD analysis within the time span between 1980 and 2012. We identify treatment firm-pairs as those that are likely to experience an increase in common ownership linkages between the two entities due to the institution mergers. Specifically, we require that (1) one party of the supply chain is held (>1% of outstanding shares) by one of the merging institutions during the quarter immediately before the merger announcement date and (2) the other party of the supply chain is held (>1% of outstanding shares) by the other party of the supply chain during the same quarter.[6] In particular, some treatment pairs may be commonly held by one of the merging institutions before the merger. These pairs are also considered "treated" because they will also experience positive exogenous shocks on the sum of shares held by common investors. For example, if a supplier–customer pair was commonly held by investor A while only the supplier was held by investor B prior to the merger, the supplier firm will experience an increase in its shares held across common investors postmerger, as its holdings from investors A and B are combined.

For control pairs, we require that one party of the supply chain is held (>1% of outstanding shares) by one of the merging institutions, while the other party of the supply chain is not held by the other merging institution. Our definition of the control pairs ensures that managerial skills or investment styles of the merging institutions that potentially relate to portfolio firms' innovation prospects are the same between control pairs and treatment pairs.

We choose to study a symmetric 7-year window around the event year (i.e., 3 years before and 3 years after the institution merger plus the year of the merger).[7] We employ a multivariate DiD approach to compare the change in innovation specificity for treatment pairs with that for control pairs, with the following regression:

$$\begin{aligned} Innovation_Specificity_{i,j,k,t+1} = {} & \alpha_1 Treat \times Post + \alpha_{12} Post + \alpha_3 Treat + \beta X_{i,t} \\ & + \gamma Y_{j,t} + \delta Z_{i,j,t} + \mu_{i,j,k} + \upsilon_t + \varepsilon_{i,j,k,t+1}, \end{aligned} \tag{2}$$

where i, j, k, and t denote supplier, customer, merger event, and year, respectively. *Treat* is a dummy that equals one for treatment pairs and zero for control pairs. *Post* is a dummy that equals one after the merger date and zero before the merger date. $Innovation_Specificity_{i,j,k,t+1}$ is the outcome variable of interest defined the same as in Equation (1), while $X_{i,t}$, $Y_{j,t}$, and $Z_{i,j,t}$ are supplier-, customer-, and pairwise-level control variables defined the same as in Equation (1). Year fixed effects (υ_t) are included to capture the time trend of innovation activities and possible fluctuation of overall economic conditions. In most specifications, we include *merger* × *pair* fixed effect which captures time-varying common trends across mergers and time-invariant characteristics across firm-pairs within merger. In particular, a firm-pair can

[6] Using the holding information in the quarter immediately before the merger announcement date ensures that we only use ex ante information to define our treatment and control firm-pairs. A key advantage of relying on ex ante information is to alleviate concern that actual holding of the institution after merger may be driven by innovation prospects of the supplier–customer firms (He & Huang, 2017).

[7] Results are similar if we use a 9-year window around the event year, and can be provided upon request.

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serve as either treatment or control in our sample for different merger events. The *merger* × *pair* fixed effects ensure that our specification identifies treatment effect through variation in innovation specificity for the same firm-pair in a given merger. We also include specifications with firm-pair fixed effects and merger fixed effects separately. We cluster standard errors by institution merger to address the possibility that firm-pairs in the same institution merger might be correlated.

We start with the DiD analysis to test whether institution mergers increase the level of CIO for treatment firms relative to control groups, that is, the first-stage analysis. In columns (1) and (2) of Table 4, Panel A, the coefficients of *Treat* × *Post* are both positive and significant at 5% level, suggesting that institution mergers indeed increase the level of CIO for treatment firms relative to control groups. These results confirm the validity of the setting.

Next, we conduct the DiD analysis using Equation (2) and tabulate the results in Table 4, Panel B. Across all specifications, the coefficients of *Treat* × *Post* are positive and significant, suggesting that suppliers who become commonly held with their customers (or receive more common ownership linkages) due to financial institution mergers conduct more relationship-specific innovation investments toward their commonly held customers, relative to the control group. In terms of economic magnitude, for the *TotalCiter* measure, the coefficients across columns (1)–(3) indicate that suppliers in the treatment pairs, relative to the control pairs, on average file 2.210 to 2.561 more patents that cite their commonly held customers', amounting to 29.4–34.0% of the standard deviation. Columns (4)–(6) indicate that suppliers cite 2.169 to 2.416 more of their commonly held customers' patents following a positive shock to their common ownership linkages, amounting to 21.4–24.9% of the standard deviation. In addition, Figure 1 shows the dynamics of the coefficients of *TotalCiter* and *TotalCited*. Using different regression specifications, we confirm that suppliers' investment specificity toward their commonly held customers only increases after the merger events, satisfying the parallel trend assumption. Overall, the DiD results corroborate our main hypothesis and provide causal evidence that common ownership along the supply chain facilitates suppliers to conduct more relationship-specific innovations toward their customers.

3.4 | Robustness tests

There could be a concern that the effects of patenting behavior might take longer than 1 year to manifest. To address this concern, we extend our baseline results to include supplier innovation outcomes in year $t + 3$ in Table 5. Panel A presents the results of CIO on the patent amount as well as suppliers' investment specificity for the OLS sample. Panel B presents the results on suppliers' investment specificity for the DiD sample. The results are similar and consistent with those in Tables 3 and 4, indicating that the effect of CIO persists long enough for suppliers' relationship-specific innovation investments to materialize.

Recent research demonstrates that the traditional staggered DID approach, which uses already-treated firms, never treated, and not-yet-treated firms as the control group, can produce biased estimates in the heterogeneous treatment effects (Baker et al., 2022). To mitigate this concern, we follow the recent econometrics literature and conduct several robustness tests. Table A1 reports the result and shows that our results are robust to a number of alternative specifications including the Callaway and Sant'Anna's (2021) estimators, Goodman-Bacon's (2021) decomposition, and Sun and Abraham's (2021) event study diagnosis.

4 | CHANNEL INVESTIGATION

4.1 | Hold-up problems mitigation channel: Asset redeployability

Our results from the OLS and DiD analyses both indicate that CIO along the supply chain promotes more relationship-specific innovation by suppliers toward their commonly held customers. In this section, we investigate

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TABLE 4 Difference-in-differences (DiD) test of common ownership on supplier innovation specificity.

Panel A: First-stage DiD results of financial institution mergers on common institutional ownership

	(1)	(2)
	NumCross	*TotalCrossOwn*
Treat × *Post*	0.112***	0.009**
	[0.039]	[0.004]
Post	−0.024	−0.002
	[0.028]	[0.002]
Treat	−0.029	−0.002
	[0.024]	[0.002]
Controls	Yes	Yes
Year FE	Yes	Yes
Merger × Firm-Pair FE	No	No
Merger FE	Yes	Yes
Firm-Pair FE	Yes	Yes
Observations	4887	4887
Adj. R^2	0.602	0.633

Panel B: DiD results of financial institution mergers on suppliers' investment specificity

	(1)	(2)	(3)	(4)	(5)	(6)
	TotalCiter			*TotalCited*		
Treat × *Post*	2.561***	2.432***	2.210***	2.416***	2.221***	2.069**
	[0.714]	[0.655]	[0.779]	[0.674]	[0.694]	[0.793]
Post	−0.334*	−0.292	−0.426**	−0.479	−0.474	−0.577**
	[0.175]	[0.178]	[0.175]	[0.291]	[0.288]	[0.271]
Treat			−0.802**			−0.416
			[0.345]			[0.507]
Controls	No	Yes	Yes	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Merger × Firm-Pair FE	Yes	Yes	No	Yes	Yes	No
Merger FE	No	No	Yes	No	No	Yes
Firm-Pair FE	No	No	Yes	No	No	Yes
Observations	4,887	4,887	4,887	4,887	4,887	4,887
Adj. R-squared	0.893	0.896	0.888	0.905	0.905	0.901

Notes: This table examines the DiD test on the relation between supplier innovation specificity and pairwise common institutional ownership by exploiting financial institution mergers. Panel A reports the first-stage results effect of DiD on common institutional ownerships. We define a pair as a treatment if the supplier is held (>1% of outstanding shares) by one of the merging institutions during the quarter immediately before the merger announcement date and the customer is held (>1% of outstanding shares) by the other party to the merger during the same premerger quarter. A pair is a control if the supplier is held by the same institution and the customer is not held by the other party to the merger, and vice versa. *Treat* is a dummy variable that equals one if a pair is a treatment and zero if it is a control. *Post* is a dummy that equals one for the postevent period and zero for the preevent period. Dependent variables are two different measures of common institutional ownership, *NumCross* and *TotalCrossOwn*. Panel B reports the results of DiD regressions on supplier innovation specificity. Dependent variables are two different measures of supplier's investment specificity, *TotalCiter* and *TotalCited*, as defined in Appendix A. Supplier level, customer level, and pairwise control variables are suppressed and defined in Appendix A. Standard errors, reported in brackets, are clustered at institution merger levels. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

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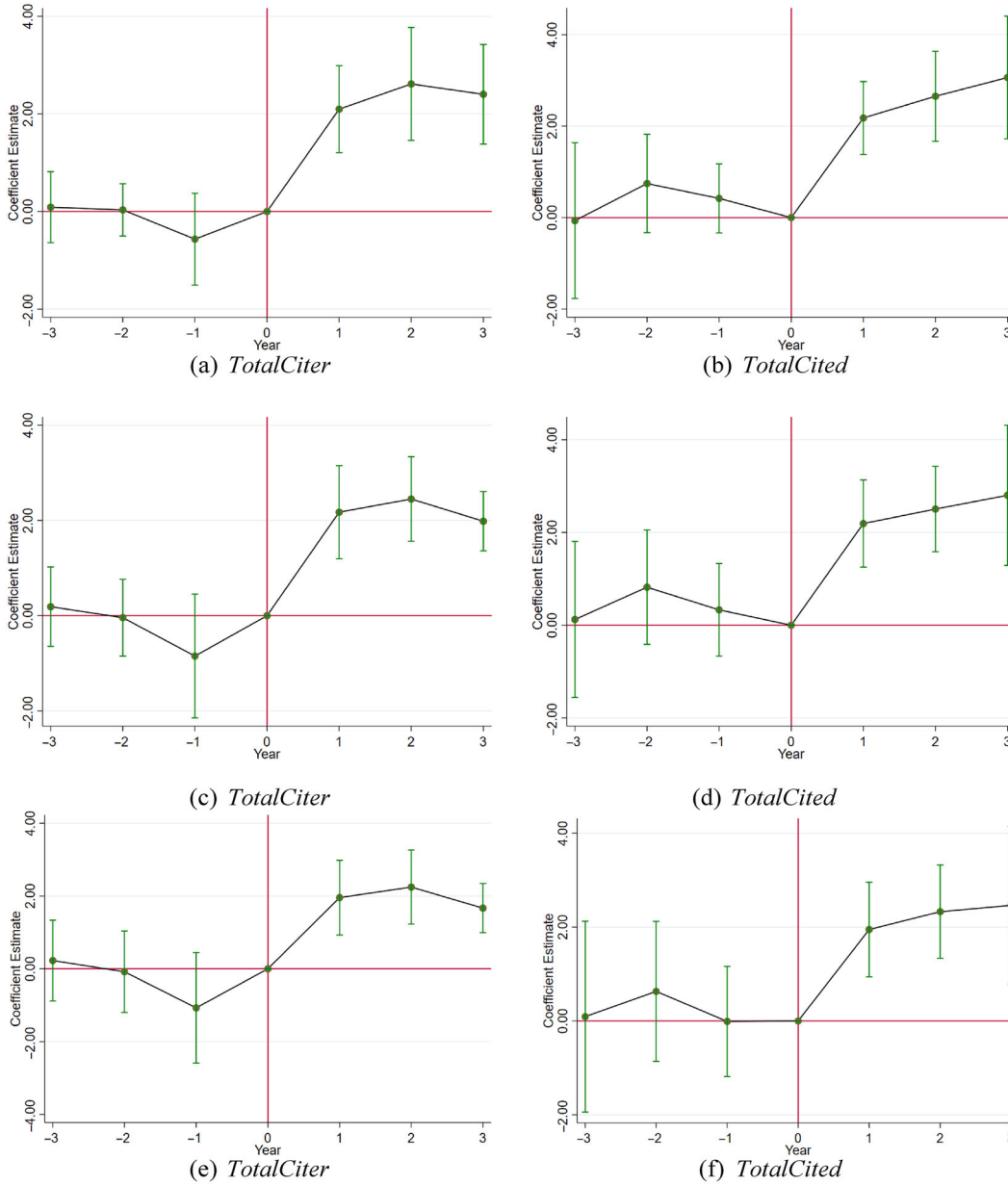

FIGURE 1 Diff-in-diff dynamics.
[Color figure can be viewed at wileyonlinelibrary.com]
Notes: The black solid lines show diff-in-diff year-by-year coefficient estimates for *TotalCiter* and *TotalCited*, which are defined in Table A1. Financial institution merger event-years are omitted. The vertical lines represent 95% confidence intervals. Panels (a) and (b) are for specification with Year and Merger × Firm-pair fixed effects; Panels (c) and (d) are for specification with Year, Merger × Firm-pair fixed effects and all control variables; and Panels (e) and (f) are for specification with Year, Merger, Firm-pair fixed effects and all control variables.

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TABLE 5 Robustness: Using innovation outcomes 3 years forward.

Panel A: OLS regression results						
	(1)	(2)	(3)	(4)	(5)	(6)
	$ln(Patent+1)_{t+3}$		$TotalCiter_{t+3}$		$TotalCited_{t+3}$	
NumCross	0.020**		0.477***		0.646***	
	[0.009]		[0.180]		[0.200]	
TotalCrossOwn		0.143		5.682		7.686*
		[0.191]		[3.544]		[4.308]
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Pair FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	22,874	22,874	5525	5525	5525	5525
Adj. R^2	0.744	0.744	0.640	0.638	0.628	0.626

Panel B: DiD results		
	(1)	(2)
	$TotalCiter_{t+3}$	$TotalCited_{t+3}$
Treat × *Post*	1.457**	1.880**
	[0.662]	[0.694]
Post	−0.710***	−0.755**
	[0.220]	[0.306]
Treat	−0.563*	−0.548
	[0.326]	[0.509]
Controls	Yes	Yes
Year FE	Yes	Yes
Merger × Firm-Pair FE	No	No
Merger FE	Yes	Yes
Firm-Pair FE	Yes	Yes
Observations	3972	3972
Adj. R^2	0.903	0.915

Notes: This table examines the dynamic effects of relationship between patenting outcomes and the level of pairwise common institutional ownership lagged by 3 years. Panel A represents the OLS regression results of patent number and suppliers' investment specificity. Panel B uses the financial institution merger setting as in Table 4 and reports the difference-in-differences (DiD) results around merger events results on suppliers' investment specificity. The main explanatory variables are two different measures of common institutional ownership, *NumCross* and *TotalCrossOwn*, as defined in Appendix A. Supplier level, customer level, and pairwise control variables are suppressed and defined in Appendix A. Standard errors, reported in brackets, are clustered at supplier–customer pair levels. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

the heterogeneous patterns in the DiD effects to provide more evidence supporting that suppliers increase relationship-specific innovation to mitigate the hold-up concern toward their customers. If the increased innovation specificity indeed results from mitigation of hold-up problems, then we should see the effect be stronger for suppliers ex ante faced with severer hold-up problems. We capture the severity of hold-up problems faced by suppliers from two dimensions. First, we consider the flexibility of suppliers' assets. We follow Kim and Kung (2017) and use asset redeployability (i.e., *Redeploy*) to capture whether a firm is more flexible to reallocate its assets. Firms with more

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TABLE 6 Heterogeneous difference-in-differences (DiD) effects by asset redeployability.

	(1)	(2)	(3)	(4)
	TotalCiter		*TotalCited*	
Treat × *Post* × *Redeploy*	−62.362***	−57.214***	−48.621***	−43.350***
	[10.783]	[12.697]	[9.370]	[10.534]
Treat × *Post*	25.309***	23.037***	20.154***	17.831***
	[4.105]	[4.904]	[3.448]	[3.991]
Treat × *Redeploy*	42.775**	21.827***	59.230***	13.793**
	[15.923]	[7.099]	[21.240]	[6.492]
Post × *Redeploy*	4.789	3.707	11.082*	8.444
	[5.002]	[4.908]	[6.315]	[5.981]
Redeploy	1.055	2.133	−7.317	−3.919
	[3.658]	[4.143]	[5.653]	[5.981]
Post	−2.050	−1.768	−4.570*	−3.677
	[1.801]	[1.770]	[2.345]	[2.238]
Treat		−8.807***		−5.494**
		[2.530]		[2.304]
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Merger × Firm-Pair FE	Yes	No	Yes	No
Merger FE	No	Yes	No	Yes
Firm-Pair FE	No	Yes	No	Yes
Observations	4887	4887	4887	4887
Adj. R^2	0.899	0.890	0.906	0.901

Notes: This table examines the heterogeneous difference-in-differences (DiD) effects of pairwise common institutional ownership on supplier innovation specificity by exploiting different levels of supplier asset redeployability prior to the merger event, proxy by asset redeployability from Kim and Kung (2017). Supplier level, customer level, and pairwise control variables are suppressed and defined in Appendix A. Standard errors, reported in brackets, are clustered at institution merger levels. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

redeployable assets exhibit higher recovery rates and trade more actively in secondary markets (He & Huang, 2017). Thus, suppliers with higher asset redeployability should concern less about being held up by their customers, as they can more easily sell their assets at a lower cost. We empirically test this hypothesis in a Difference in difference in differences (DDD) model by adding interaction terms of *Redeploy* × *Treat*, *Redeploy* × *Post*, and *Redeploy* × *Treat* × *Post*. We use firms' asset redeployability 1 year before the merger to show the channel effect.

We tabulate the regression results in Table 6. Across columns (1)–(4), the coefficients of *Treat* × *Post* × *Redeploy* are negative and significant (at the 1% level), indicating that for suppliers with higher asset redeployability prior to the merger, the increase in innovation specificity after being commonly held with their major customers is lower. Economically, compared to firms with lower asset redeployability, firms with higher redeployability see a decrease of 4.005 (= 0.07*57.214) *TotalCiter* for each standard deviation increase in their asset redeployability. Similarly, compared to firms with lower asset redeployability, firms with higher redeployability see a decrease of 3.035 (= 0.07*43.350) *TotalCited* for each standard deviation increase in redeployability. The results support the hypothesis that for suppliers faced with more severe hold-up problems, common ownership with their customers is particularly relevant in promoting innovation specificity.

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TABLE 7 Heterogeneous difference-in-differences (DiD) effects by bargaining power.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Supplier-HHI				*Customer-HHI*			
	TotalCiter		*TotalCited*		*TotalCiter*		*TotalCited*	
Treat × Post × HHI	−13.276**	−13.278**	−11.841**	−12.831**	16.129***	14.751***	10.791***	9.283**
	[5.646]	[5.379]	[5.001]	[4.620]	[2.645]	[3.050]	[3.263]	[3.517]
Treat × Post	5.780**	5.665**	5.173**	5.385**	−3.739***	−3.567***	−1.916	−1.573
	[2.228]	[2.330]	[2.105]	[2.167]	[0.714]	[0.840]	[1.432]	[1.404]
Treat × HHI	12.474**	7.877***	7.373	6.785***	−18.051*	−7.834***	−13.409*	−5.456
	[5.349]	[2.827]	[4.831]	[2.032]	[9.425]	[2.793]	[7.545]	[3.351]
Post × HHI	−1.222	−0.624	−2.849	−1.467	−0.126	−0.217	−2.408	−1.980
	[1.716]	[1.598]	[1.944]	[1.840]	[1.006]	[0.979]	[2.194]	[1.916]
HHI	3.287*	5.281**	4.726**	5.724**	8.667***	5.137**	9.813**	6.132**
	[1.816]	[1.934]	[1.813]	[2.170]	[2.643]	[2.306]	[3.534]	[2.499]
Post	0.088	−0.226	0.389	−0.125	−0.241	−0.331	0.433	0.175
	[0.604]	[0.533]	[0.773]	[0.687]	[0.514]	[0.440]	[1.071]	[0.893]
Treat		−2.689***		−2.005**		2.206*		1.673
		[0.962]		[0.862]		[1.181]		[1.576]
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger × Firm-Pair FE	Yes	No	Yes	No	No	Yes	No	No
Merger FE	No	Yes	No	Yes	Yes	No	Yes	Yes
Firm-Pair FE	No	Yes	No	Yes	Yes	No	Yes	Yes
Observations	4887	4887	4887	4887	4887	4887	4887	4887
Adj. R^2	0.897	0.889	0.906	0.901	0.898	0.889	0.906	0.901

Notes: This table examines the heterogeneous DiD effects of pairwise common institutional ownership on supplier innovation specificity by exploiting different levels of bargaining power of both supplier and customer prior to the merger event, proxy by Herfindahl–Hirschman Index (HHI), with similar regression specification as in Table 6. Supplier level, customer level, and pairwise control variables are suppressed and defined in Appendix A. Standard errors, reported in brackets, are clustered at institution merger levels. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

4.2 | Hold-up problems mitigation channel: Bargaining power

The second dimension of firm heterogeneity we explore is suppliers' bargaining power. Following prior studies, we use industry HHI to proxy for the suppliers' and customers' relative bargaining power. For suppliers operating in industries with higher HHI, an indicator for less competition within the industry, they should have higher bargaining power relative to their customers, thus be less concerned with being held up. If customers have relatively large bargaining power, on the other hand, suppliers should be more concerned with being held up in the future. Similar to previous analysis, we incorporate supplier's and customer's HHI 1 year before the merger event into the DiD regression. The results are reported in Table 7. The first four columns study how suppliers' bargaining power interacts with the effects of common ownership. The coefficients on *Treat* × *Post* × *Sup_HHI* suggest that for suppliers with higher bargaining power, the increase in innovation specificity after being commonly held with their major customers is lower. Economically, compared to suppliers with lower HHI, suppliers with higher HHI see a decrease of 3.000 (= 0.226*13.278) *TotalCiter* for each standard deviation increase in suppliers' HHI, and a decrease of 2.900 (= 0.226*12.831) *TotalCited* for each standard deviation increase in suppliers' HHI. Columns (5)–(8) study how customers' bargaining power interacts with the effects of common ownership. The results are just the opposite, with the main variable of interest, *Treat* × *Post* × *Cust_HHI* being positive and significant.

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TABLE 8 Difference-in-differences (DiD) test of common ownership on customer innovation specificity.

	(1)	(2)	(3)	(4)	(5)	(6)
	Cust_TotalCiter			*Cust_TotalCited*		
Treat × *Post*	−0.751	−0.688	−0.641	−1.900**	−1.980**	−1.795**
	[0.588]	[0.555]	[0.578]	[0.728]	[0.724]	[0.805]
Post	0.338	0.316	0.049	0.219	0.161	−0.217
	[0.358]	[0.333]	[0.443]	[0.482]	[0.457]	[0.658]
Treat			0.208			0.712
			[0.922]			[1.142]
Controls	No	Yes	Yes	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Merger × Firm-Pair FE	Yes	Yes	No	Yes	Yes	No
Merger FE	No	No	Yes	No	No	Yes
Firm-Pair FE	No	No	Yes	No	No	Yes
Observations	4478	4478	4478	4478	4478	4478
Adj. R^2	0.838	0.841	0.824	0.852	0.854	0.821

Notes: This table examines the DiD test on the relation between customer innovation specificity and pairwise common institutional ownership by exploiting financial institution mergers using the same regression specification as in Table 4, with the dependent variables measuring customers' innovation specificity. Supplier level, customer level, and pairwise control variables are suppressed and defined in Appendix A. Standard errors, reported in brackets, are clustered at institution merger levels. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

The heterogeneity analysis corroborates with our main findings and lends further support to the hold-up mitigation channel. Under circumstances where suppliers are faced with severer hold-up problems—either with lower asset redeployability or lower bargaining power relative to their major customers—common ownership along the supply chain exerts a larger impact on promoting innovation specificity.

4.3 | Alternative explanations: Knowledge spillover

Our next test considers an alternative explanation—knowledge spillover. If our main results are driven by knowledge spillover, then we should observe not only that suppliers conduct more specialized innovation toward their customers, but customers' innovation being more specialized to their suppliers. In other words, the effect of knowledge spillover should be two-way. We might observe both suppliers' and customers' innovations becoming more aligned with each other during the exchanges of ideas and know-hows. We test this explanation by looking at customers' innovation specificity. Specifically, we define *Cust_TotalCiter* as the number of customer patents that cite its suppliers' patents in an application year, and *Cust_TotalCited* as the number of supplier patents that are cited by the customer in an application year. The results are reported in Table 8. Our main variable of interest, *Treat* × *Post*, is statistically insignificant across columns (1)–(3), while it is significantly negative in columns (4)–(6). This suggests that customers cite less of their suppliers' patents after being commonly held due to institution mergers, which is against the prediction of the knowledge spillover channel.

4.4 | Alternative explanations: Collaboration

We next consider firm collaboration as another possible channel driving our main results, as common investors may promote R&D collaborations between their commonly held suppliers and customers, thus achieving more

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TABLE 9 Difference-in-differences (DiD) test of common ownership on innovation collaboration.

	(1)	(2)	(3)
	Alliances		
Treat × *Post*	−0.016**	−0.016**	−0.017**
	[0.007]	[0.006]	[0.006]
Post	0.006	0.006	0.007
	[0.006]	[0.006]	[0.006]
Treat			0.006
			[0.006]
Controls	No	Yes	Yes
Year FE	Yes	Yes	Yes
Merger × Firm-Pair FE	Yes	Yes	No
Merger FE	No	No	Yes
Firm-Pair FE	No	No	Yes
Observations	4887	4887	4887
Adj. R^2	0.067	0.069	0.150

Notes: This table examines the DiD test on the relation between innovation collaboration and pairwise common institutional ownership by exploiting financial institution mergers using the same regression specification as in Table 4, with the dependent variable *Alliances*, a dummy indicator for whether the supplier and customer establish a joint venture or strategic alliance in a year. Supplier level, customer level, and pairwise control variables are suppressed and defined in Appendix A. Standard errors, reported in brackets, are clustered at institution merger levels. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

relationship-specific innovations without necessarily addressing the hold-up concerns between the two parties. To test for this potential mechanism, we consider an explicit form of R&D collaboration, that is, joint ventures and strategic alliances established by the two trading partners. Prior studies have found that joint ventures and strategic alliances are important organizational forms for two distinct entities to conduct innovation collaboration (Robinson, 2008; Seru, 2014). If common ownership promotes innovation specificity through facilitating innovation collaboration, we expect to see the probability of establishing a joint venture and strategic alliance to be higher after the two trading partners become commonly held. We collect information on all the joint ventures and strategic alliances between the suppliers and customers in our sample from the SDC Platinum Database. We estimate a linear probability model using the DiD framework as in Equation (2) and tabulate the results in Table 9. Across columns (1)–(3), *Treat* × *Post* is negative and significant, suggesting that suppliers and customers are less likely to form joint ventures and strategic alliances after being commonly held. This evidence helps us rule out firm collaboration as an alternative channel. One possible explanation for the decrease in innovation collaboration could be the substitution effect, as the increase in suppliers' innovation specificity toward their commonly held customers spares the customers from either tailoring their technology to fit into their suppliers' more generalized products (the reason why we see less innovation specificity toward suppliers by customers), or forming joint ventures and strategic alliances that aim to research on know-hows to meet customers' specialized needs. This is consistent with the argument viewing vertical joint ventures as alternatives to simple contracts, and firms in buyer–supplier transactions only choose joint ventures when hold-up costs are high (Johnson & Houston, 2000).

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TABLE 10 Difference-in-differences (DiD) test of common ownership on supply chain values.

	(1) *Sup_TobinQ*	(2) *Cust_TobinQ*	(3) *Pair_TobinQ*
Treat × *Post*	0.028	0.209**	0.195**
	[0.104]	[0.090]	[0.084]
Post	−0.123***	−0.020	−0.012
	[0.036]	[0.032]	[0.031]
Treat	−0.074	−0.061	−0.061
	[0.065]	[0.061]	[0.058]
Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Firm-Pair FE	Yes	Yes	Yes
Observations	4887	4887	4887
Adj. R^2	0.732	0.811	0.804

Notes: This table examines the DiD test on the relation between supply chain valuations and pairwise common institutional ownership by exploiting financial institution mergers using the same regression specification as in Table 4. The dependent variables are the supplier-, customer-, and pairwise Tobin's Q, *Sup_TobinQ*, *Cust_TobinQ*, *Pair_TobinQ* at year $t + 1$. We remove the market-to-book control variables for both supplier and customer as they are highly correlated to the value of Tobin's *Q* in the regressions. Supplier level, customer level, and pairwise control variables are suppressed and defined in Appendix A. Standard errors, reported in brackets, are clustered at institution merger levels. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

5 | WELFARE IMPLICATIONS

Our results so far have provided evidence that CIO along the supply chain promotes relationship-specific innovation investments of suppliers toward their commonly held customers by mitigating the hold-up problem. In this section, we explore the welfare implications on the investors from such effect.

Institutional investors aim to maximize the risk-adjusted returns of its portfolio. A substantial body of literature has argued that shareholders with diversified portfolios aim to maximize portfolio profits rather than individual firm profits, and as a result, portfolio diversification can mitigate market frictions (Admati et al., 1994; Clayton & Jorgensen, 2005; Hansen & Lott, 1996; Rubin, 2006). Hansen and Lott's (1996) model suggests that individual firms may impose various types of externalities—pecuniary or nonpecuniary, vertical or horizontal—on one another, and diversification helps internalize these externalities to maximize portfolio returns. Therefore, we predict that an institution holding stakes in both customers and suppliers could align incentives between the two parties and reduce the risk of opportunistic behaviors, resulting in an improved coordination with higher overall returns.

To empirically test this prediction, we examine whether CIO involvement leads to higher valuations (measured by Tobin's Q) of suppliers, customers, and supplier–customer pairs using a similar DiD framework. Specifically, the joint valuation of a supplier–customer pair at a given year is defined as the combined book value of assets minus the book value of common equity plus the market value of common equity of both supplier and customer, then scaled by the combined book value of assets of both supplier and customer. Table 10 reports the results. We find that the coefficients of *Treat* × *Post* across columns (1)–(3) are positive and statistically significant. In column (2), the coefficient of *Treat* × *Post* is 0.209, indicating a 10.6% (= 0.209/1.965) increase in Tobin's *Q* for treated customers compared to the control group. In column (3), the coefficient of *Treat* × *Post* is 0.195, indicating a 9.9% (= 0.195/1.972) increase in the combined value of the treated pairs, relative to the control pairs.

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TABLE 11 Investor types.

	(1)	(2)	(3)	(4)
	TotalCiter		*TotalCited*	
NumCross_TRA	0.266*		0.440**	
	[0.161]		[0.213]	
NumCross_QIX	0.341***		0.586***	
	[0.121]		[0.160]	
NumCross_DED	0.507**		0.742**	
	[0.248]		[0.349]	
TotalCrossOwn_TRA		3.650		6.804
		[3.757]		[4.880]
TotalCrossOwn_QIX		3.722		9.137***
		[2.348]		[3.041]
TotalCrossOwn_DED		0.716		0.692
		[3.193]		[3.802]
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm-Pair FE	Yes	Yes	Yes	Yes
Observations	8783	8783	8783	8783
Adj. R^2	0.652	0.650	0.656	0.653

Notes: This table examines the relation between supplier innovation activities and type of common owned investors. We follow Bushee's (1998, 2001) classification of institutional investors and decompose our measures of CIO, *NumCross* and *TotalCrossOwn*, into three variables indicating "dedicated" (DED), "quasi-indexer" (QIX), and "transient" (TRA) investors. Standard errors, reported in brackets, are clustered at institution merger levels. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

The results on firm value suggest that common ownership of a supplier–customer pair significantly creates value for investors, particularly in the form of customer valuations. Our findings are consistent with prior studies, which also report increased value effects of CIO involvement (e.g., Freeman, 2023; He & Huang, 2017). Combined with our main findings that CIO involvement may mitigate supply chain hold-up problems for suppliers, CIO involvement along the supply chain could be Pareto improving, leading to beneficial outcomes for all three parties.

We next investigate the mechanism through which common institutional investors exert influence on their portfolio companies' operations. First, we examine the types of institutions based on Bushee's (1998, 2001) classification to determine whether "dedicated" (DED), "quasi-indexer" (QIX), or "transient" (TRA) institutions that commonly hold supplier–customer pairs lead to the cooperation along the supply chain.[8] According to Bushee's definition, DED investors actively select assets to own for the long term, while QIX and TRA investors hold a diversified portfolio of assets in a more passive manner. Specifically, we decompose *NumCross* into three variables (i.e., *NumCross_DED*, *NumCross_QIX*, and *NumCross_TRA*) representing the number of unique common investors of each type, and decompose *TotalCrossOwn* into three variables (i.e., *TotalCrossOwn_DED*, *TotalCrossOwn_QIX*, and *TotalCrossOwn_TRA*) representing the sum of supplier shares held by each type of common investors, divided by the supplier's total shares outstanding. We then conduct horserace regressions using Equation (1). Table 11 reports the results. We find that CIO of all three types has significantly positive effect on supplier's investment specificity. Particularly, in columns (1) and (3), the coef-

[8] We are grateful to Brian Bushee for providing these data on his website.

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ficients of *NumCross_DED* are larger than those of *NumCross_TRA* and *NumCross_QIX*, suggesting that active investors have a stronger effect than passive investors in facilitating cooperation along the supply chain.

Next, we examine the "cross-blockholders" who hold at least 3% or 5% of both suppliers and customers, instead of the 1% threshold used in our baseline analysis. In untabulated results, we find that the impact of CIO on investment specificity using the 3% or 5% cutoff is economically similar to our baseline results but not statistically significant. This lack of statistical significance is due to the rarity of such cases in our sample. The average number of unique common investors holding both the supplier and customer with holdings larger than 1% of total shares outstanding is 2.107, but this sharply decreases to 0.517 for the 3% cutoff and 0.179 for the 5% cutoff. This observation is consistent with Freeman (2023), who also found that such cases are extremely limited.

6 | CONCLUSION

In this paper, we analyze supplier investment specificity toward their commonly held customers via patent cross-citations, and show that vertical CIO can mitigate hold-up problems along the supply chain. We find that when commonly held with their major customers, suppliers conduct more specialized R&D and produce more specialized patents, more so when the common ownership networks are wider and deeper. We establish causality using a DiD approach based on a quasi-natural experiment of financial institution mergers. Our results are robust to a battery of robustness tests. Furthermore, we investigate the differential effects of vertical CIO by suppliers' ex ante hold-up concern severity and provide evidence on hold-up concerns mitigation as the underlying mechanism driving the increase in suppliers' innovation specificity. We also provide evidence that our main findings are stronger in active institutions rather than passive institutions. As the last step of our analysis, we show that common ownership of a supplier–customer pair creates value for investors, particularly benefiting customer valuations.

Our paper not only tries to tackle hold-up problems by offering a new solution which is independent of actions taken by the two parties of the business relationship, but is also among the first to extend the common ownership literature to study its vertical effects along the supply chain. Our paper sheds new light on the hold-up concerns mitigation effect of common ownership on suppliers' relationship-specific innovation investments toward customers. In view of this positive effect, our study has important policy implications given the recent debates on imposing tighter regulations on financial institutions' cross-holdings due to the widely documented anticompetition effect (on firms competing in the same industries).[9] Our study provides new evidence on the efficiency implications of common ownership that future regulatory discussions should take into consideration.

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[9] "Common ownership of shares faces regulatory scrutiny" (*Financial Times*, January 22, 2019). https://www.ft.com/content/59325462-fe57-11e8-aebf-99e208d3e521.

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APPENDIX A: VARIABLE DEFINITIONS

Variable name	Description
NumCross	The number of unique common investors holding both the supplier and customer (with holdings larger than 1% of total shares outstanding).
TotalCrossOwn	The sum of supplier shares held across common investors, divided by the supplier's total share outstanding (holdings larger than 1% of total shares outstanding).
Sup_R&D	Supplier R&D expenditures divided by the book value of total assets.
Sup_MB	Supplier market value of total assets divided by the book value of total assets.
Sup_Size	Supplier natural logarithmic of the book value of total assets.
Sup_Leverage	Supplier total debt divided by the book value of total assets.
Sup_ROA	Supplier operating income divided by the book value of total assets.
Sup_Tangibility	Supplier net property, plant, and equipment (PPE) divided by the book value of total assets.
Sup_Capex	Supplier capital expenditures divided by the book value of total assets.
Sup_InstOwn	Supplier total shares held by institutional investors divided by total shares outstanding.
Cust_MB	Customer market value of total assets divided by the book value of total assets.
Cust_Size	Customer natural logarithmic of the book value of total assets.
Cust_Leverage	Customer total debt divided by the book value of total assets.
Cust_ROA	Customer operating income divided by the book value of total assets.

(Continues)

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Variable name	Description
Cust_Tangibility	Customer net PPE divided by the book value of total assets.
Cust_Capex	Customer capital expenditures divided by the book value of total assets.
Cust_InstOwn	Customer total shares held by institutional investors divided by total shares outstanding.
SalesFraction	Supplier's sales to each major customer divided by total sales.
ln(*Patents*+1)	Natural logarithmic of one plus the number of patents filed and eventually granted in each application year.
TotalCiter	The number of supplier patents that cite its customer's patents in an application year.
TotalCited	The number of customer patents that cited by its supplier in an application year.
Cust_TotalCiter	The number of customer patents that cite its supplier's patents in an application year for difference-in-differences (DiD) sample.
Cust_TotalCiter	The number of supplier patents that cited by its customer in an application year for DiD sample.
Redeploy	The redeployability score from Kim and Kung (2017). Asset redeployability score is calculated as the value-weighted average of the industry-level redeployability index across the business segments. Industry-level redeployability is calculated as the value-weighted average of the asset-level redeployability scores from the Bureau of Economic Analysis (BEA) 1997 capital flow table. Asset-level redeployability scores is calculated as the sum of market capitalization of all public firms in industries that use the asset over the sum of market capitalization across all public firms in the 123 industries in the BEA 1997 capital flow table.
Sup_HHI	The sum of squared market shares in sales of the supplier's industry. Industry is defined using four-digit Standard Industrial Classification (SIC) code.
Cust_HHI	The sum of squared market shares in sales of the customer's industry. Industry is defined using four-digit SIC code.
Sup_TobinQ	Suppliers' the book value of assets (at) minus the book value of common equity (ceq) plus the market value of common equity (csho*prcc_f), then scaled by the book value of assets (at).
Cust_TobinQ	Customers' the book value of assets (at) minus the book value of common equity (ceq) plus the market value of common equity (csho*prcc_f), then scaled by the book value of assets (at).
Pair_TobinQ	Combined the book value of assets (at) minus the book value of common equity (ceq) plus the market value of common equity (csho*prcc_f) of both supplier and customer, then scaled by the combined the book value of assets (at) of both supplier and customer.
Alliances	A dummy variable for whether the supplier and customer establish a joint venture or strategic alliance in a year.

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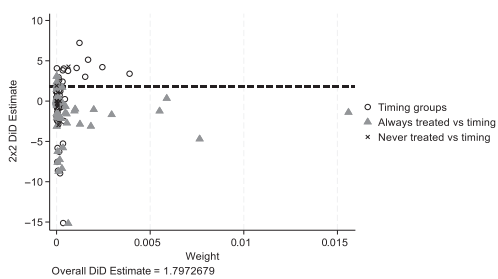
TABLE A1 Robustness: Alternative estimators to account for heterogeneous treatment effects.

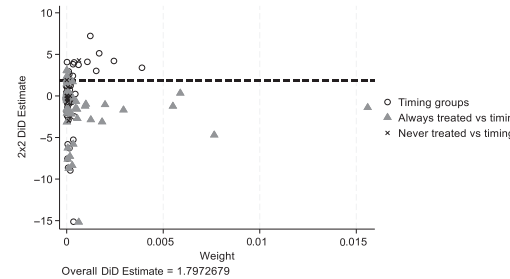
Panel A: Using Callaway and Sant'Anna's (2021) estimators

	(1)	(2)	(3)	(4)
	TotalCiter		*TotalCited*	
	NumCross	*TotalCrossOwn*	*NumCross*	*TotalCrossOwn*
ATT	3.090***	2.653***	2.495***	1.924***
	[0.868]	[0.779]	[0.535]	[0.599]
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm-Pair FE	Yes	Yes	Yes	Yes
Observations	4322	4043	4322	4043
Cluster at	Merger event	Merger event	Merger event	Merger event

Panel B: Using Goodman-Bacon's (2021) decomposition

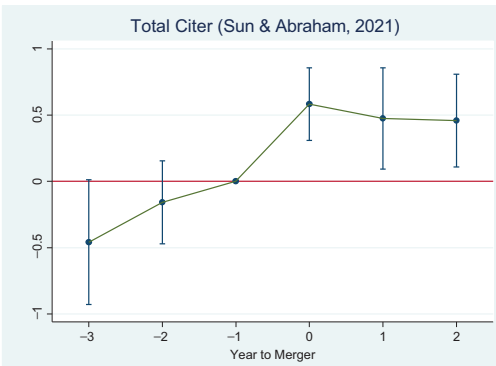


(a) *TotalCiter*

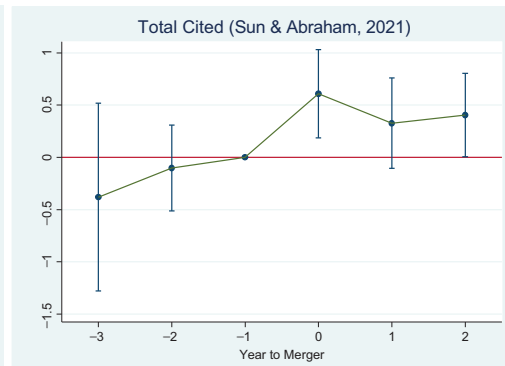


(b) *TotalCited*

Panel C. Using Sun and Abraham's (2021) event study method



(a) *TotalCiter*



(b) *TotalCited*

Note: This table reports the results of alternative estimators to account for heterogeneous treatment effects in our staggered difference-in-differences (DID) setting. Panel A reports the results using Callaway and Sant'Anna's (2021) estimators by the Stata command "csdid." Panel B presents the Goodman-Bacon's (2021) estimators using the Stata command "bacondecomp." Panel C presents Sun and Abraham's (2021) estimator using the Stata command "eventstudyinteract." Supplier level, customer level, and pairwise control variables are suppressed and defined in Appendix A. Standard errors, reported in brackets, are clustered at the institution merger level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

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